Course title : **CSE3009**

Course title : **NoSQL Databases**

Module : **0**

Topic : **0**

Zero Hour

Objectives

This session will give the knowledge about

- Course syllabus
- Objectives
- Course outcome
- CO's Mapping with PO's and PEO’s
- Couse plan

Objectives

1. Explores the origins of NoSQL databases and their characteristics that distinguish them from traditional RDBMS.

2. Explains the architectures and common features of the main types of NoSQL databases (key-value stores, document databases, column-family stores, graph databases)

3. Discusses the criteria that decision makers should consider when choosing between relational and non-relational databases and techniques for selecting the NoSQL database that best addresses specific use cases.

Expected Outcome

1. Explain the detailed architecture, define objects, load data, query data and performance tune NoSQL databases

2. Define NoSQL, its characteristics, history and primary benefits using NoSQL Databases.

3. Define the major types of NoSQL databases including a primary use case and advantages/disadvantages of each type.

4. Analyze semi-structured data and choose an appropriate storage structure

Course Outcomes

Course Outcomes	Course Outcome Statement	PO's / PEO’s
CO1	Explain the detailed architecture, define objects, load data, query data and performance tune NoSQL databases	PO1, PO2, PO3, PO4, PO5
CO2	Define NoSQL, its characteristics, history and primary benefits using NoSQL Databases.	PO1, PO2, PO3, PO4, PO5
CO3	Define the major types of NoSQL databases including a primary use case and advantages/disadvantages of each type.	PO1, PO2, PO3, PO4, PO5, PO9, PO11
CO4	Analyze semi-structured data and choose an appropriate storage structure	PO1, PO2, PO3, PO4, PO5, PO9, PO11

Programme Outcomes (PO) (CSE-All Specialization)

After successful completion of the program a student is expected to have abilities to:

- **PO1. Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- **PO2. Problem analysis:** Identify, formulate, research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

Programme Outcomes (PO) (CSE-All Specialization)

- **PO3. Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- **PO4. Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

Programme Outcomes (PO) (CSE-All Specialization)

- **PO5. Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
- **PO6. The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues, and the consequent responsibilities relevant to the professional engineering practice.
- **PO7. Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

Programme Outcomes (PO) (CSE-All Specialization)

- **PO8.Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- **PO9. Individual and teamwork:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- **PO10. Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

Programme Outcomes (PO) (CSE-All Specialization)

- **PO11. Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one’s own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- **PO12. Life-long learning:** Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Syllabus

Module No. 1	Introduction To NoSQL Concepts	hours: 06 hrs
Data base revolutions: First generation, second generation, third generation, Managing Transactions and Data Integrity, ACID and BASE for reliable database transactions, Speeding performance by strategic use of RAM, SSD, and disk, Achieving horizontal scalability with database sharding, Brewer's CAP theorem.		
Module No. 2	**NoSQL Data Architecture Patterns**	**hours: 08 hrs**
NoSQL Data model: Aggregate Models, NoSQL system ways to handle big data problems, Moving Queries to data, not data to the query, hash rings to distribute the data on clusters, replication to scale reads, Database distributed queries to data nodes.		

Syllabus

Module No. 3	Key –Value Data Stores	hours: 08 hrs
From array to key –value databases, Essential features of key – value Databases, Properties of keys, Characteristics of Values, Key-Value Database Data Modeling Terms, Key-Value Architecture and implementation Terms, Limitations of Key- Value Databases, Case Study: Key-Value Databases for Mobile Application Configuration.		
Module No. 4	Document Oriented Database	hours: 07 hrs
Document, Collection, Naming, CRUD operation, querying, indexing, Replication, Sharding, Consistency Implementation: Distributed consistency, Eventual Consistency, Capped Collection, Case studies: document oriented database: MongoDB and/or Cassandra		

Syllabus

Module No. 5	Columnar Data Model	hours: 08 hrs
Data warehousing schemas: Comparison of columnar and row-oriented storage, Column-store Architectures: C-Store and Vector-Wise, Column-store internals and, Inserts/updates/deletes, Indexing. Advanced techniques: Vectorized Processing, Compression, Write penalty, Joins, Group-by, Aggregation and Arithmetic Operations, Case Studies		
Module No. 6	Data Modeling with Graph	hours: 08 hrs
Comparison of Relational and Graph Modeling, Property Graph Model Graph Analytics: Link analysis algorithm- Web as a graph, PageRank- Page rank computation, Topic specific page rank (Page Ranking Computation techniques: iterative processing, Random walk distribution) Querying Graphs: Introduction to Cypher, case study: Building a Graph Database Application.		

Syllabus

Text Books:

1. Guy Harrison, “Next Generation database: NoSQL and Big data”, A Press, 2016
2. Sullivan, Dan. "NoSQL for mere mortals.", Addison-Wesley Professional, 2015.
3. Dan McCreary and Ann Kelly. "Making sense of NoSQL." Shelter Island: Manning, 2014.

Syllabus

Reference Books:

1. Daniel Abadi, Peter Boncz, Stavros Harizopoulos, Stratos Idreaos, Samuel Madden, “The Design and Implementation of Modern Column-Oriented Database Systems”, Now Publishers,2013
2. Perkins, Luc, Eric Redmond, and Jim Wilson. "Seven databases in seven weeks: a guide to modern databases and the NoSQL movement." Pragmatic Bookshelf, 2018.
3. Tiwari, Shashank. "Professional nosql." John Wiley & Sons, 2011.

Mark Configuration

CAT-1: Module 1 and Module 2 (18th Aug 2025) – 15%

CAT-2: Module 3 and Module 4 (6th Oct 2025) – 15%

FAT: All Modules (21st Dec 2025) – 40%

Non-CAT Assessment (30%):

Digital Assignment-1: Marks

Class Notes: Marks

QUIZ-1: Marks

Certification:

Attendance: Marks

Summary

We have discussed about

- Course syllabus
- Objectives
- Course outcome
- CO's Mapping with PO's and PEO’s
- Couse plan